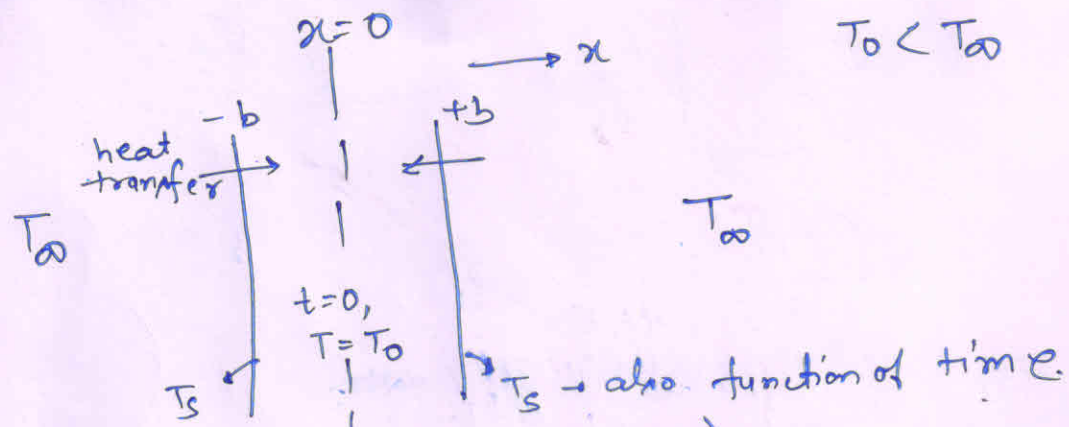


desg 4
Q.2



unidirectional heat transfer (only conduction)

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

I.C. $t=0, T=T_0$

B.C. $x=0, \frac{\partial T}{\partial x}=0$, from symmetry.
 $\uparrow$ time dependant

$$x=\pm b, -k \frac{\partial T}{\partial x} = h(T_s - T_0)$$

Non dimensionalization

$$\theta = \frac{T_\infty - T}{T_\infty - T_0} \quad \tau = \frac{t}{t^*}$$

$$\eta = \frac{x}{b} \quad t^* = ?$$

$t^* = \frac{b^2}{\alpha}$
 (from order of magnitude analysis)

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \Rightarrow \frac{T_\infty - T_0}{t^*} \sim \alpha \frac{(T_\infty - T_0)}{b^2} \Rightarrow t^* \sim \frac{b^2}{\alpha}$$

$$\frac{\partial \left(\frac{T_\infty - T}{T_\infty - T_0} \right)}{\partial \tau} = \frac{\alpha \frac{\partial^2 \left(\frac{T_\infty - T}{T_\infty - T_0} \right)}{\partial \eta^2}}{b^2}$$

$$\Rightarrow \frac{\partial \theta}{\partial \tau} = \frac{\alpha}{b^2} \frac{\partial^2 \theta}{\partial \eta^2}$$

$$\left(\frac{b^2}{\alpha} \frac{1}{b^2} \right) = 1$$

$$\left(\frac{\partial \theta}{\partial \tau} \right) = \frac{\partial^2 \theta}{\partial \eta^2} \quad \text{--- (a)}$$

I.C. $\tau=0, \theta = \frac{T_\infty - T_0}{T_\infty - T_0} = 1$

B.C. $x=0 \Rightarrow \eta=0, \frac{\partial T}{\partial x}=0 \Rightarrow \frac{\partial \theta}{\partial \eta}=0$

$x=\pm b \Rightarrow \eta=\pm 1$

$$(-) k \frac{\partial T}{\partial x} \Big|_{\eta=+1} = h(T_s - T_0)$$

$$k \frac{\partial (T)}{\partial x} \Big|_{\eta=-1} = h(T_s - T_0)$$

$$\frac{k \left(\frac{T_\infty - T}{T_\infty - T_0} \right)}{b} \Big|_{\eta=+1} = -h \frac{(T_\infty - T_s)}{(T_\infty - T_0)}$$

$$\frac{k}{b} \frac{\partial \theta}{\partial \eta} \Big|_{\eta=+1} = -h \theta$$

B.C $\eta = +1$

$$\left. \frac{k}{b} \frac{\partial \theta}{\partial \eta} \right|_{\eta=+1} = -h \theta$$

$$\Rightarrow \left. \frac{\partial \theta}{\partial \eta} \right|_{\eta=+1} = - \left(\frac{hb}{k} \right) \theta$$

$$\Rightarrow \left. \frac{\partial \theta}{\partial \eta} \right|_{\eta=+1} = - \text{Bi} \theta \quad \left(\text{Biot Number} \right)$$

$x = +b$
 $\eta = +1$

(d)

Let us revisit eqn (a) & apply separation of variables.

$$\theta = f(\eta) g(\tau)$$

$$\frac{\partial \theta}{\partial \tau} = \frac{\partial^2 \theta}{\partial \eta^2} \Rightarrow f \frac{dg}{d\tau} = g \frac{d^2 f}{d\eta^2}$$

$$\Rightarrow \frac{dg}{g d\tau} = \frac{1}{f} \frac{d^2 f}{d\eta^2} = -c^2$$

$$g = A \exp(-c^2 \tau) \quad \text{--- (b)}$$

$$\frac{d^2 f}{d\eta^2} + c^2 f = 0$$

$$f = B \cos(c\eta) + C \sin(c\eta)$$

as θ is symmetric.
(from physics of problem
temp profile is symmetric)

$$f = B \cos c\eta \quad \text{--- (c)}$$

$$\Rightarrow f(+\eta) = f(-\eta)$$

$$\Rightarrow f \text{ must be } B \cos(c\eta)$$

from eqn (d) (B.C)

$$\left. \frac{\partial \theta}{\partial \eta} \right|_{\eta=+1} = -(Bi) \theta \Big|_{\eta=+1}$$

$$\Rightarrow \cancel{g} \left. \frac{\partial f}{\partial \eta} \right|_{\eta=+1} = (Bi) \cancel{g} f \Big|_{\eta=+1}$$

$$\Rightarrow \left. \frac{\partial f}{\partial \eta} \right|_{\eta=+1} = (Bi) f \Big|_{\eta=+1} \quad \text{--- (e)}$$

from eqn (c)

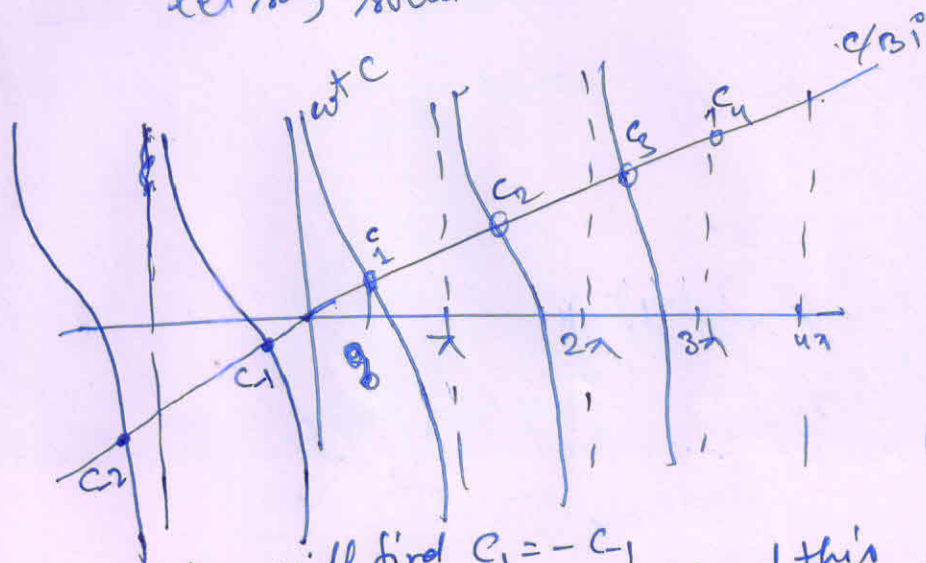
$$\left. \frac{\partial (B \cos(\eta))}{\partial \eta} \right|_{\eta=+1} = (Bi) B \cos(\eta) \Big|_{\eta=+1}$$

$$\cancel{B \sin(c)} \cdot B c (\sin c) = (Bi) B (\cos c)$$

$$c = \frac{Bi}{\tan c} = (Bi) (\cot c)$$

✓ solution of $C = Bi \cot C$

let say solution are $C = C_n \quad n=1, 2, 3, 4$



one has to solve numerically

① ② ③ ④
point of intersection
or solution

you will find $C_1 = -C_1$
(in class I missed this discussion)

C_n

$$\theta = \sum g_n f_n = \sum (A_n \exp(-c_n^2 z)) (B_n \cos(c_n \eta))$$

$$\theta = \sum_{n=-\infty}^{\infty} P_n \exp(-c_n^2 z) \cos(c_n \eta)$$

final solution

$$\text{let say } f_n = P_n \cos(c_n \eta)$$

we know f_n satisfies

$$\frac{d^2 f_n}{d\eta^2} + c_n^2 f_n = 0$$

$$\text{multiplying } f_m = P_m \cos(c_m \eta)$$

$$f_m \frac{d^2 f_n}{d\eta^2} + c_n^2 f_n f_m = 0$$

integrating both sides: $\eta=0 \rightarrow 1$

$$\int_0^1 f_m \frac{d^2 f_n}{d\eta^2} d\eta + \int_0^1 c_n^2 f_n f_m d\eta = 0$$

$$\Rightarrow f_m \frac{df_n}{d\eta} \Big|_0^1 - \int_0^1 \frac{df_m}{d\eta} \frac{df_n}{d\eta} d\eta + \int_0^1 c_n^2 f_n f_m d\eta = 0$$

at $\eta=1$

$$\frac{df_n}{d\eta} = (-B_i) f_n$$

form $\exp(\theta)$

$$\eta=0 \quad \frac{df_n}{d\eta} = 0$$

$$f_m \frac{df_n}{d\eta} \Big|_{\eta=1} - f_m \frac{df_n}{d\eta} \Big|_{\eta=0} + \int_0^1 c_n^2 f_n f_m d\eta = 0$$

$$-B_i f_n f_m \Big|_{\eta=1} + \int_0^1 \frac{df_m}{d\eta} \frac{df_n}{d\eta} d\eta + \int_0^1 c_n^2 f_n f_m d\eta = 0$$

$$-Bi f_n f_m \Big|_{\eta=1} + \int_0^1 \frac{df_m}{d\eta} \frac{df_n}{d\eta} d\eta + \int_0^1 c_n^2 f_n f_m d\eta = 0 \quad \text{--- (f)}$$

swapping n & m

$$-Bi f_n f_m \Big|_{\eta=1} + \int_0^1 \frac{df_n}{d\eta} \frac{df_m}{d\eta} d\eta + \int_0^1 c_m^2 f_n f_m d\eta = 0 \quad \text{--- (g)}$$

$$\underline{\text{Eqn (f)} - \text{(g)}}$$

$$(c_m^2 - c_n^2) \int_0^1 f_n f_m d\eta = 0$$

$$c_m = c_n \Rightarrow m = n$$

$$\text{or } m \neq n$$

$$\int_0^1 f_n f_m d\eta = 0$$

verifying the final solution

$$\theta = \sum_{n=-\infty}^{\infty} P_n e^{-c_n^2 \tau} \cos(c_n \eta)$$

Now let's say c_n is solution of $c \tan c = Bi^*$
 $\Rightarrow c_n \tan c_n = Bi^*$

★ during class I missed this discussion

we find that $(-c_n)$ also satisfies the above eqn

$$(-c_n)(-\tan c_n) = (-c_n) \tan(-c_n) = Bi^*$$

$$\Rightarrow \text{for } c_n \text{ \& } c_{-n} = -c_n$$

we get

$$\text{for } c = c_{-n} \Rightarrow P_n e^{-c_n^2 \tau} \cos(c_n \eta) = P_{-n} e^{-(-c_n)^2 \tau} \cos(-c_n \eta)$$

$$\Rightarrow P_{-n} e^{-c_n^2 \tau} \cos(c_n \eta) = P_{-n} e^{-c_n^2 \tau} \cos(-c_n \eta)$$

for $c = c_n$ $P_n = P_n e^{-c_n^2 \tau} \cos(c_n \eta)$

thus as we add c_n & $-c_n$ terms we will get

$$\underbrace{(P_{-n} + P_n)}_{Q_n} e^{-c_n^2 \tau} \cos(c_n \eta)$$

of course $n=0 \Rightarrow c_n$ will also be there

so we have now

$$\theta = \sum_{n=0}^{\infty} Q_n e^{-c_n^2 \tau} \cos(c_n \eta)$$

for $n=0 \Rightarrow \infty$

$$c_1 = 1.3138$$

$$c_2 = 4.0336$$

$$c_3 = 6.9026$$

$$c_4 = 9.8921$$

you can also check

$$n \rightarrow -\infty \rightarrow 0$$

$$c_{-1} = -1.3138$$

$$c_{-2} = -4.0336$$

too large exp. term will $\rightarrow 0$

so no need to go further

check on any values that $c_n = -c_n$

applying 1.c $\tau=0, \theta=1$

$$\Rightarrow 1 = \sum_{n=0}^{\infty} Q_n \cos(c_n \eta)$$

multiply by $\cos(c_m \eta)$ and integrate from 0 to π

$$\int_0^{\pi} \cos(c_m \eta) d\eta = \int_0^{\pi} \sum_{n=0}^{\infty} Q_n \underbrace{\cos(c_n \eta)}_{f_n} \underbrace{(\cos(c_m \eta))}_{f_m} d\eta$$

orthogonality $\int_0^{\pi} f_n f_m d\eta = 0 \quad m \neq n$

$$\int_0^1 \cos(c_m \eta) d\eta = \sum_{n=0}^{\infty} Q_n \int_0^1 \cos(c_n \eta) \cos(c_m \eta) d\eta$$

$$\Rightarrow \int_0^1 \cos(c_m \eta) d\eta = Q_m \int_0^1 (\cos(c_m \eta))^2 d\eta$$

↳ except for $m=n$
all other terms will
vanish due to orthogonality.

$$\Rightarrow \frac{\sin(c_m \eta)}{c_m} \Big|_0^1 = Q_m \int_0^1 (1 + \cos(2c_m \eta)) d\eta$$

$$\Rightarrow \frac{\sin c_m}{c_m} = (Q_m) \left[\eta \Big|_0^1 + \frac{\sin(2c_m \eta)}{2c_m} \Big|_0^1 \right]$$

$$\Rightarrow \frac{\sin c_m}{c_m} = Q_m \left[1 + \frac{\sin 2c_m}{2c_m} \right]$$

$$Q_m = \frac{2 \sin c_m}{2 c_m + \sin 2 c_m} = \frac{2 \sin c_m}{2 c_m + 2 \sin c_m \cos c_m}$$

$$Q_m = \boxed{\frac{1}{c_m + \cos c_m}}$$

$$\theta = \sum_{n=0}^{\infty} \left(\frac{1}{c_n + \cos c_n} \right) \exp(-c_n^2 \tau) \cos(c_n \eta)$$

(in case I
made a
mistake in
integration,
kindly correct)

$$\frac{T_{\infty} - T}{T_{\infty} - T_0} = \sum_{n=0}^{\infty} \left(\frac{1}{c_n + \cos c_n} \right) \exp\left(-c_n^2 \left(\frac{t}{B^2/\alpha}\right)\right) \cos\left(c_n \frac{x}{R}\right)$$

(where c_n are roots of $c \tan c = Bi$)